Goo counts

Cornstarch Goo

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little Wonderer pours 46 spoonfuls of powder, then 27 more. How many spoonfuls in all?

46 + 27 = _____

Show your thinking.

2. A resting spoon sinks 52 millimetres, but a smacked spoon sinks only 18. How much deeper is the resting spoon?

52 − 18 = _____

Show your thinking.

3. A drawing shows 38 sliding grains and 45 jammed grains. How many grains in all?

38 + 45 = _____

Show your thinking.

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4. A squeezed ball shows 63 tiny cracks, and 28 melt away first. How many cracks are left?

63 − 28 = _____

Show your thinking.

5. Big Wonderer counts 19 slow drips, and the little one counts 24. How many drips in all?

19 + 24 = _____

Show your thinking.

6. A beach shows 72 footprints, and waves wash away 36. How many footprints are left?

72 − 36 = _____

Show your thinking.

Soft, then hard

One bowl of goo. A slow mitten sinks right in, but a fast tap feels as hard as a table.

Slow sinks, fast stops

A resting spoon sinks slowly. A smacked spoon stops on top. Water never acts like that.

Grains in a crowd

Tiny grains float in water. Slow pushes let them slide. Fast pushes jam them into chains.

Dry, then shiny

Squeeze a goo ball, and it looks dry and cracked. Stop, and it melts into a shiny puddle.

A slow ribbon

The goo is one crowd of grains. Gentle pushes let them slide; quick pushes jam them.

Pale sand by the sea

A step pushes wet sand grains apart, so water sinks in and the top looks dry. How long can a jam last?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

Cornstarch Goo

Explain a goo clue in your words

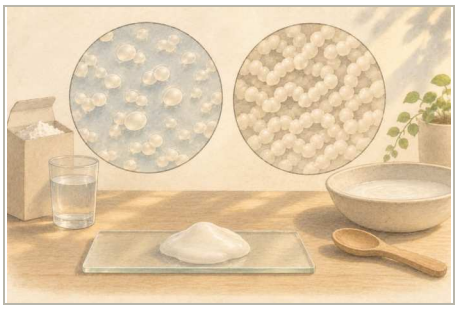
Cornstarch Goo · Curiosity day

Name _____ Date _____

Choose one picture. Say your idea first, then draw or write it.



Slow sinks, fast stops



Grains in a crowd



Dry, then shiny

Draw two bowls of goo: one with a spoon sunk deep, one with a spoon resting on a dent.

slow · fast · grain · push · because · maybe

The smacked spoon stops on top because … what does a fast push do?

The grains jam when I push fast because … where can the water go?

The squeezed ball looks dry because … where do the grains poke out?

Does your sentence tell what you noticed and use because or maybe to explain?



The Spoon They Forgot

Little Wonderer mixed up a bowl full of goo,
two scoops of white powder, one splash of water too.
Push slow, and your mitten sinks down with a sigh.
Push fast, and it's hard as a rock. My, oh my!

"It's grains," said the big one, "too tiny to see,
all slipping on water as slick as can be.
Go gently, they glide past each other with ease.
Go quickly, they jam like a crowd in a squeeze."

So the little one rolled it, round, round, in a ball,
and it looked dry and cracked, like no goo at all.
But the moment it stopped, the ball came undone
and dribbled away in a puddle of fun.

"So is it a liquid, or is it a rock?"
Big Wonderer shrugged. "It depends on the knock."
They argued till supper, both sure they were right,
while the spoon they forgot sank slowly from sight.

Cornstarch Goo

01 Why does a fast smack stop at the top?



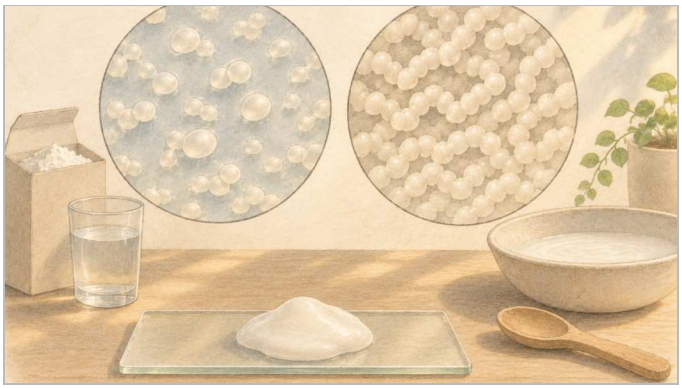
Slow sinks deep; a fast smack holds.

Look closer. Compare the deep, sunk spoon with the dull dent in the other bowl.

A little explanation. Most liquids, like water, stay just as runny however fast you push. Isaac Newton described that rule, so this goo is called non-Newtonian. A slow push lets it flow. A fast smack makes it act like a solid for a moment.

Try this. Mix about 2 scoops of cornstarch with 1 scoop of water. Rest a spoon on top, then tap the goo fast with another. Which one sinks?

02 What happens to the grains when you push fast?



Slow pushes let grains slide. Fast pushes jam them.

Look closer. Find the grains floating apart and the grains crowded into chains.

A little explanation. Each cornstarch grain is so tiny that about five fit across one hair. Water sits between them like a slippery layer. A slow push lets the water slip aside. A fast push shoves the grains together before the water can move, so they lock up. Scientists call this jamming.

Try this. Drag a spoon slowly through the goo, then quickly. Feel how much harder your arm has to work.

03 Why does squeezed goo look dry?



Squeezed goo looks dry, then melts.

Look closer. Look at the dull, cracked skin of the ball and the shiny puddle beside it.

A little explanation. Jammed grains need a little more room, so they poke up through the water at the surface. The skin turns dull and cracks. Stop squeezing, and the water covers them again, so the goo shines and melts.

Try this. Roll a lump of goo between your palms, then stop and open your hands. Watch it melt and drip.

04 Why does wet sand go pale around a foot?



A step makes wet sand look dry for a moment.

Look closer. Find the pale ring around the press mark in the wet sand.

A little explanation. A step pushes the sand grains a little apart. Water sinks into the new gaps, so the top looks dry for a moment. Osborne Reynolds explained this in 1885 and named it dilatancy. Cornstarch goo does a quick version of the same trick.

Try this. Next time you are at a beach, step on firm wet sand. Watch the pale ring appear, then lift your foot.